

pil = python imaging library

```
In [1]: from PIL import Image
```

```
In [9]: lion_image=Image.open(r'C:\Users\kraja\Downloads\lion.jpg')  
lion_image
```

Out[9]:



```
In [11]: type(lion_image)
```

Out[11]: PIL.JpegImagePlugin.JpegImageFile

```
In [13]: import numpy as np
```

```
In [14]: lion_arr=np.asarray(lion_image)
```

```
In [15]: lion_arr
```

```

Out[15]: array([[140, 158, 182],
               [140, 158, 182],
               [140, 158, 182],
               ...,
               [137, 162, 192],
               [137, 162, 192],
               [137, 162, 192]],

              [[140, 158, 182],
               [140, 158, 182],
               [140, 158, 182],
               ...,
               [138, 163, 193],
               [138, 163, 193],
               [138, 163, 193]],

              [[141, 159, 183],
               [141, 159, 183],
               [141, 159, 183],
               ...,
               [138, 163, 193],
               [138, 163, 193],
               [138, 163, 193]],

              ...,

              [[101, 86, 79],
               [102, 88, 79],
               [107, 90, 82],
               ...,
               [103, 85, 71],
               [114, 88, 73],
               [124, 95, 79]],

              [[104, 87, 80],
               [103, 86, 78],
               [106, 87, 80],
               ...,
               [105, 87, 73],
               [121, 95, 80],
               [136, 104, 89]],

              [[107, 90, 83],
               [105, 86, 79],
               [106, 87, 80],
               ...,
               [109, 90, 76],
               [128, 100, 88],
               [144, 112, 97]]], dtype=uint8)

```

```
In [16]: type(lion_arr)
```

```
Out[16]: numpy.ndarray
```

```
In [17]: import matplotlib.pyplot as plt
```

```
In [18]: plt.imshow(lion_image)
```

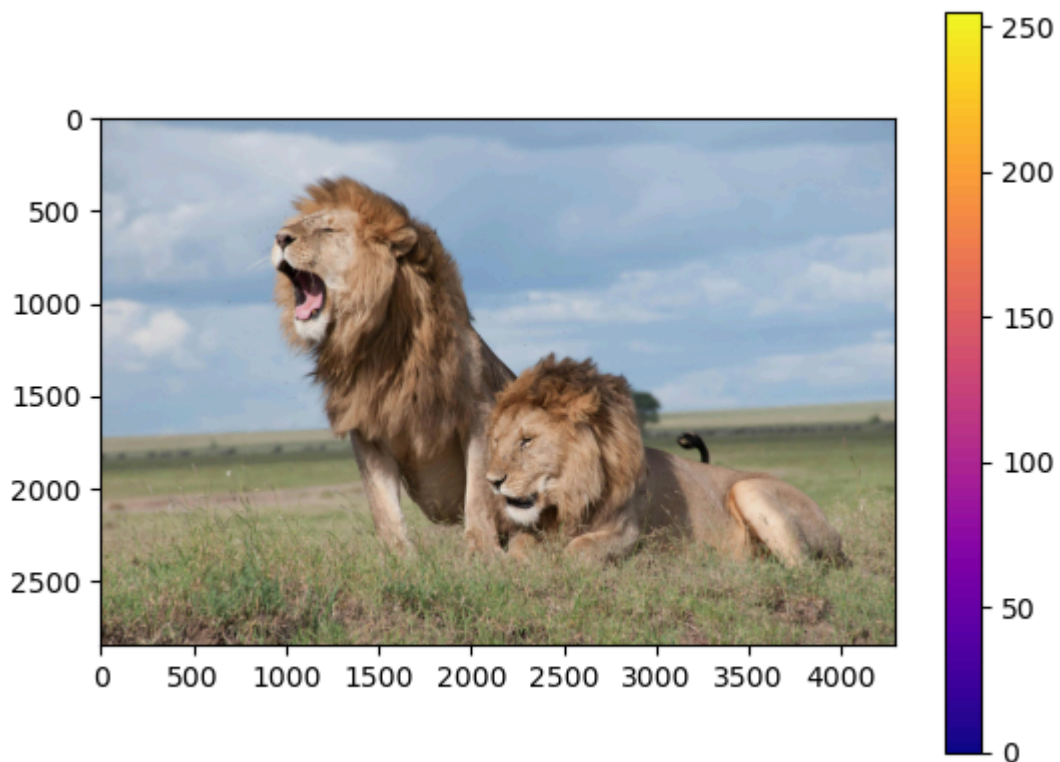
```
Out[18]: <matplotlib.image.AxesImage at 0x2ba3db10830>
```



```
In [19]: lion_arr.shape
```

```
Out[19]: (2848, 4288, 3)
```

```
In [27]: plt.imshow(lion_image, cmap='plasma')  
plt.colorbar() # Adds a legend showing the value-to-color mapping  
plt.show()
```



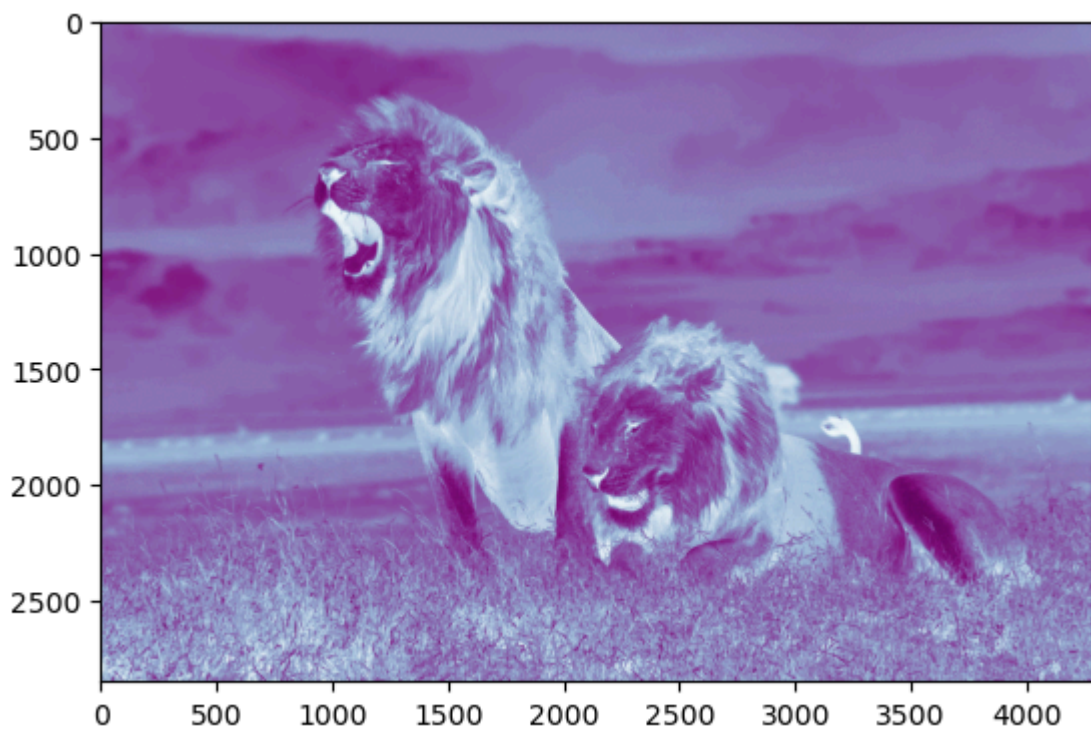
```
In [30]: plt.imshow(lion_arr[:, :, 0])
```

```
Out[30]: <matplotlib.image.AxesImage at 0x2ba4649a0d0>
```



```
In [32]: plt.imshow(lion_arr[:, :, 0], cmap='BuPu')
```

```
Out[32]: <matplotlib.image.AxesImage at 0x2ba462a9f90>
```



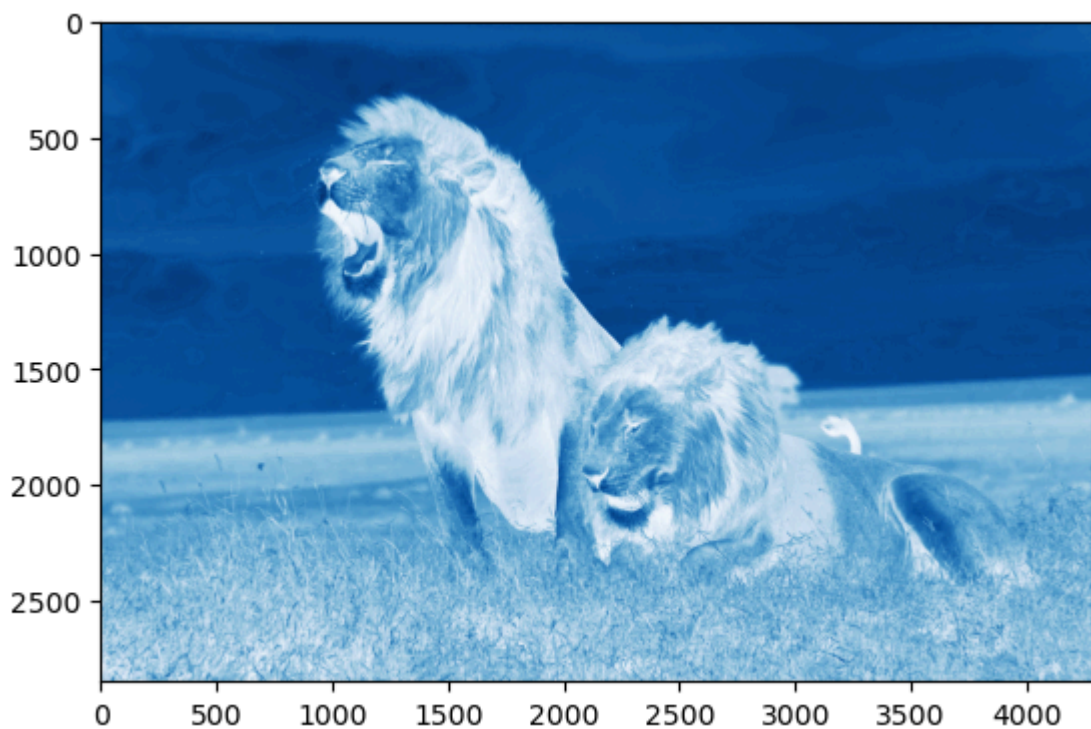
```
In [33]: plt.imshow(lion_arr[:, :, 0], cmap='Reds')
```

```
Out[33]: <matplotlib.image.AxesImage at 0x2ba46554cd0>
```




```
In [37]: plt.imshow(lion_arr[:, :, 2], cmap='Blues')
```

```
Out[37]: <matplotlib.image.AxesImage at 0x2ba46616ad0>
```



```
In [ ]:
```